

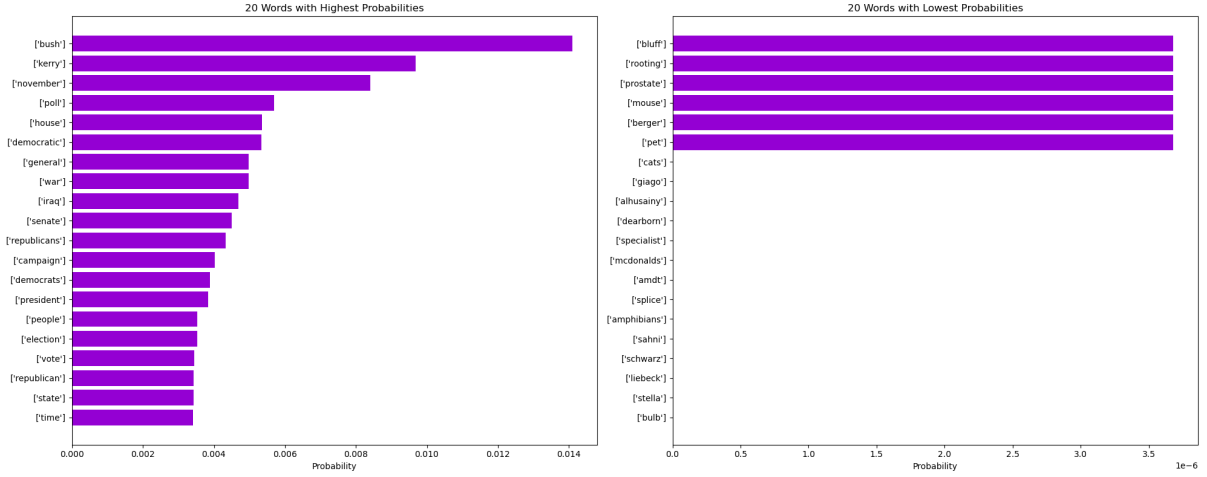


Figure 1: 20 words with highest and lowest maximum likelihood probabilities

- (a) From lecture notes, the maximum likelihood estimation (MLE) over words is $\beta_m^{ML} = c_m/N$, where c_m corresponds to the total count of the vocabulary word m and N is the total number of words in the collection.

$$p(w|\hat{\beta}) = \prod_{m=1}^M \hat{\beta}_m^{c_m} \quad (1)$$

Since the log function is concave, maximising/minimising the test set log probability is equivalent to maximising/minimising the test set probability (Eq. 1). The straightforward answer is the test set probability minimum. If the test set includes a word k not included in our training set (e.g. cats), $\hat{\beta}_k = 0$, which yields $p(w|\hat{\beta}) = 0$ when plugging in Equation 1 or $\log p(w|\hat{\beta}) = -\infty$. We can be sure this is the minimum value as 0 is the minimum value of any probability. The largest test set probability occurs when our test set includes only one appearance of the most probable word l such as:

$$\arg \max_w p(w|\hat{\beta}) = \beta_l = \arg \max_{m \in [1, M]} \beta_m \quad (2)$$

$\beta_l = 0.014 \therefore \log p(w|\hat{\beta}) = -4.27$ for the most probable word in our training set, bush.

MLE (and frequentist approaches in general) completely disregards unseen words by assigning them zero probability, as it relies exclusively on observed frequencies in training data. This zero-probability assignment to unseen events exemplifies MLE's primary weakness: overfitting to training data, which compromises the model's ability to generalize to new, unseen examples.

- (b) Bayesian inference with a symmetric Dirichlet prior with a concentration parameter α on the word probabilities β , ie $p(\beta|\alpha)$, as well as $\mathbf{c} = c_1, \dots, c_m$ observed counts of each word m , gives the following posterior:

$$p(\beta|\mathbf{w}, \alpha) \propto p(\mathbf{w}|\beta)p(\beta|\alpha) = \text{Dir}(\alpha + \mathbf{c}) \quad (3)$$

The predictive probability that the next word takes value j is:

$$\beta_j = p(w = j|\mathbf{c}) = \int p(j|\beta)p(\beta|\mathbf{c})d\beta = \mathbb{E}[\beta_j|\mathbf{c}] = \frac{\alpha + c_j}{\sum_{m=1}^M (\alpha + c_m)} = \frac{\alpha + c_j}{M\alpha + N} \quad (4)$$

Whereas MLE is a point estimate that depends exclusively on observations, Bayesian inference uses information given in the prior and solves the problem of null probabilities for unseen words. Setting $\alpha = 0$ in Equation 4 will give us the same approximation for MLE that $\beta_j = 0$ if the word j doesn't appear in the training set. Instead, if a given word doesn't appear in the training set, $c_j = 0 \therefore \forall \alpha > 0, \beta_j = \frac{\alpha}{M\alpha + N} > 0$. Irrespective of whether or not they appeared in the training dataset, $\forall j \in [1; M], \lim_{\alpha \rightarrow \infty} \beta_j = 1/M$. This observation is reinforced by Figures 1 ($\alpha = 0$) & 2, where we also notice that the order of probability remains unchanged $\forall \alpha$.

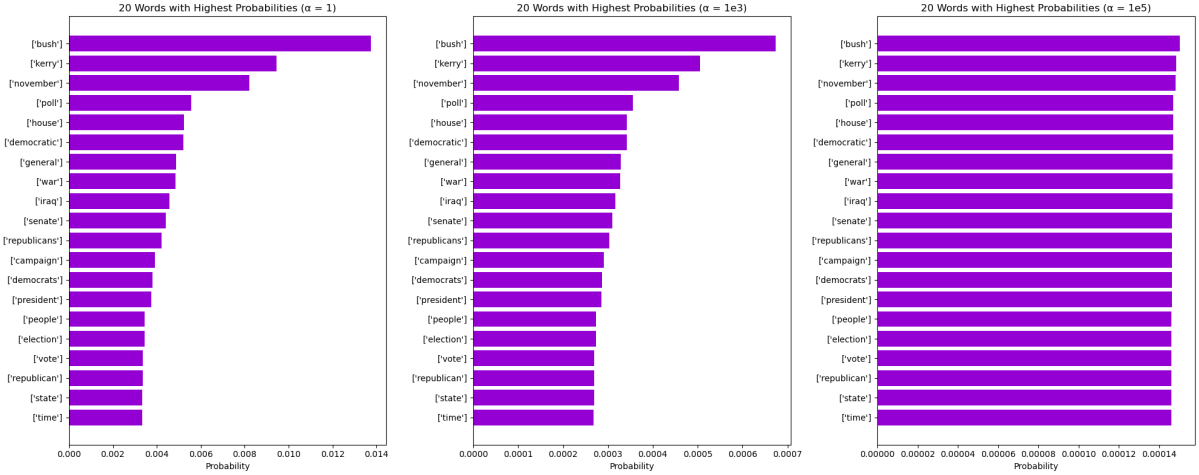


Figure 2: 20 words with highest probabilities for $\alpha = 1$, $\alpha = 10^3$ & $\alpha = 10^5$

This implies that more common words have a higher probability of being picked with lower alphas. Conversely, for a higher alpha, the distribution becomes closer to uniform: common words are less likely to be picked, opposite to rare words that are more likely to be chosen.

- (c) Using β calculated in question (b), we're able to obtain an expression for the log joint probability over the words in the document, l , by taking the log of Equation 1:

$$\log p(w|\hat{\beta}) = \sum_{m=1}^M c_m \log \hat{\beta}_m \quad (5)$$

For this calculation, we will use the categorical distribution since it takes into account the order of the words, conversely to the multinomial distribution. Then, we will compute the per-word perplexity of the document, defined as $e^{-l/n}$, where n is the number of words in the document. A perplexity of g corresponds to the uncertainty associated with a die with g sides, which generates each new word.

Document ID	Log probability	Perplexity
2001	-3,668.62	4,373.11
All documents	-1,545,980	2,683.98

Table 1: Log Probability and Perplexity of Document ID = 2001 and all documents for $\alpha = 1$

From Table 1, we observe that the log probability is lower for document id = 2001 than for all the documents combined. This result is intuitive: since each word has a probability between 0 and 1, including more factors in the probability computations yields a lower probability for smaller sequences. This idea was previously used when determining the

maximum probability of question (a), which is why the maximum probability occurs when only one word (the most probably one) appears in the document. However, the perplexity is higher for the single document 2001 than for the ensemble of all documents. A lower perplexity indicates better predictive performance, as it demonstrates the model is more certain about its predictions. This also makes sense: when considering all documents, there is less chance of having "unlikely" words. The idea is similar to the Law of Large Numbers: with more words in our set, the word distribution should come closer to the theoretical word distribution. Smaller documents have higher variances of those unlikely words and hence perplexity is higher for single documents. Different documents have varying perplexities, as some documents contain more high-probable words than others. Since Document 2001's perplexity is larger than for all documents, it has a higher proportion of more "unlikely" words than all documents have.

For a uniform multinomial, the log probability l can be computed using Equation 1 to obtain $l = N \ln(M)$, which yields a perplexity M . Perplexity in this case increases as we have more words in our set. This intuitively corresponds to having a uniform distribution of words: word frequency in the training dataset doesn't matter, as we only look if a word appeared before or not. This simplistic approach, on top of not considering word order, represents the upper bound of perplexity: models such as the categorical distribution get better results.

(d) The Gibbs sampler for Bayesian Mixture Models (BMM) involves iterative sampling:

$$\begin{aligned} p(\beta_k|w, z) &\propto \text{Dir}(\gamma + c_{mk}) \\ p(z_d = k|w_d, \theta, \beta) &\propto \theta_k p(w_d|\beta_{z_d}) \\ p(\theta|z, \alpha) &\propto \text{Dir}(c + \alpha) \end{aligned}$$

With collapsed Gibbs sampling, marginalizing over θ gives:

$$p(z_d = k|w_d, z_{-d}, \beta, \alpha) \propto p(w_d|\beta_k) \frac{\alpha + c_{-d,k}}{\sum_{j=1}^K \alpha + c_{-d,j}}$$

where latent variables become dependent rather than conditionally independent given θ .

While mixing proportions θ_k stabilize within individual chains, as evidenced by plateauing trajectories in Figure 3, this local stability does not guarantee convergence to the true posterior distribution. Different seeds lead to convergence to distinct stationary distributions at various rates, indicating the sampler identifies local modes rather than fully exploring the posterior space.

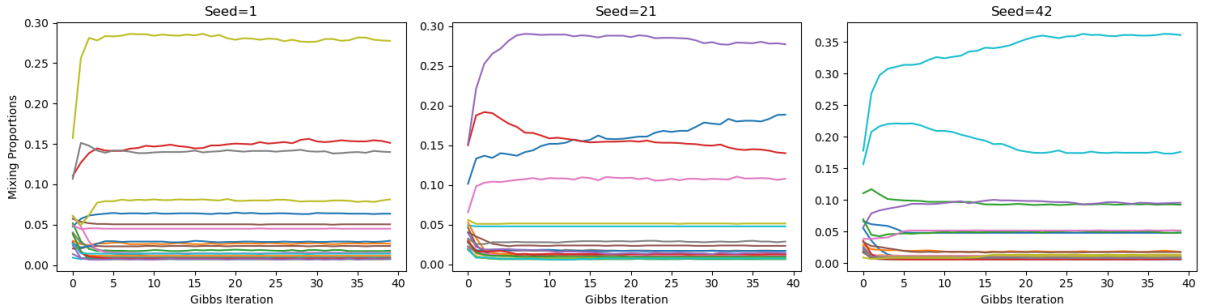


Figure 3: Mixing proportions for each topic wrt Gibbs iterations for 3 different random seeds

The multiplicative term $\frac{\alpha + c_{-d,k}}{\sum_{j=1}^K \alpha + c_{-d,j}}$ creates positive feedback - topics with higher counts $c_{-d,k}$ receive proportionally higher probability mass for new assignments. This mechanism explains why topics that accumulate early assignments tend to dominate (aka the *rich get richer* property, particularly visible for seed = 42), leading to different but similarly-structured local optima rather than thorough posterior exploration. The limitation stems from the categorical nature of z and its strong coupling with θ in the model structure, making large-scale topology changes extremely improbable under the standard Gibbs transitions despite apparent local mixing.

- (e) Latent Dirichlet Allocation (LDA) addresses a key limitation of the BMM in topic modeling. While BMM constrains all words in a document to share a single topic, LDA allows each word w_{nd} to be drawn from its own topic $z_{nd} \sim \text{Cat}(\theta_d)$, where θ_d is the document-specific topic distribution. This flexibility enables LDA to capture documents spanning multiple themes through a hierarchical generative process: each document has a topic distribution drawn from a Dirichlet prior $\theta_d \sim \text{Dir}(\alpha)$, each topic has a word distribution $\beta_k \sim \text{Dir}(\gamma)$, and words are generated by first sampling a topic then sampling from that topic’s word distribution.

From Figure 4, we observe significant variation in topic proportions across documents. Topics with larger proportions (e.g. the red line) demonstrate more focused word distributions and consequently lower entropies, as seen in Figure 5, where dominant topics stabilize at both quicker and lower entropy values. Both the topic posteriors θ_{dk}^* and word entropies stabilize after 20-30 Gibbs sweeps, suggesting 50 sweeps achieve convergence.

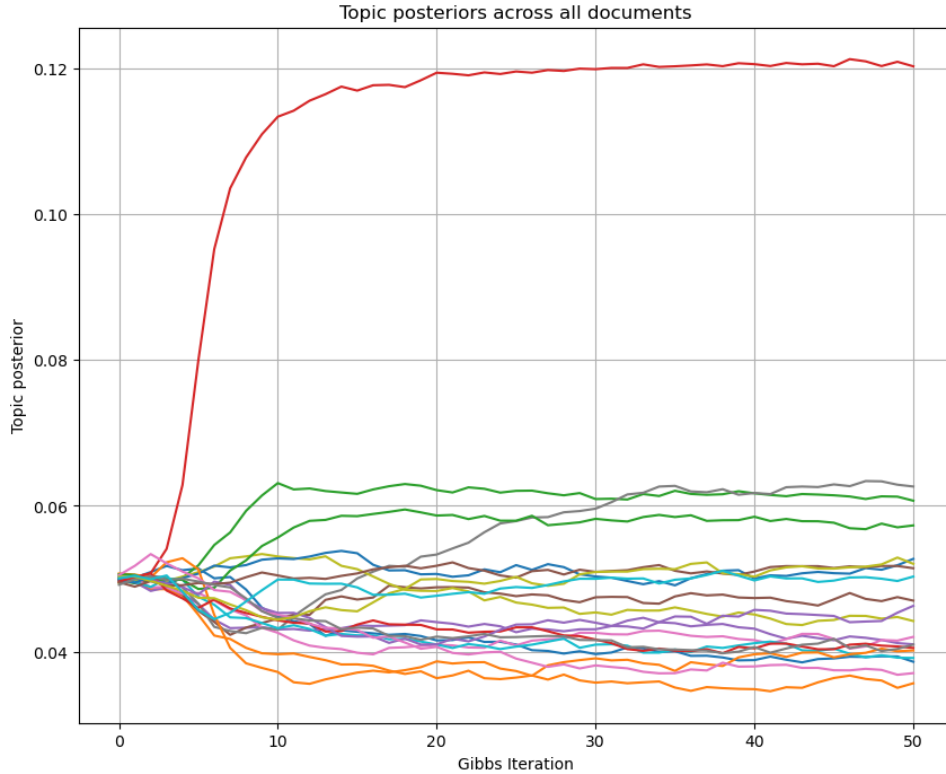


Figure 4: Topic posterior probabilities averaged over all documents wrt Gibbs iterations

The perplexity values in Table 2 quantify each model’s predictive power. LDA achieves superior performance by capturing document-specific topic mixtures. BMM’s improved perplexity over Bayesian inference demonstrates the value of multiple topics, even when restricted to one per document. Bayesian inference performs worst as it forces all docu-

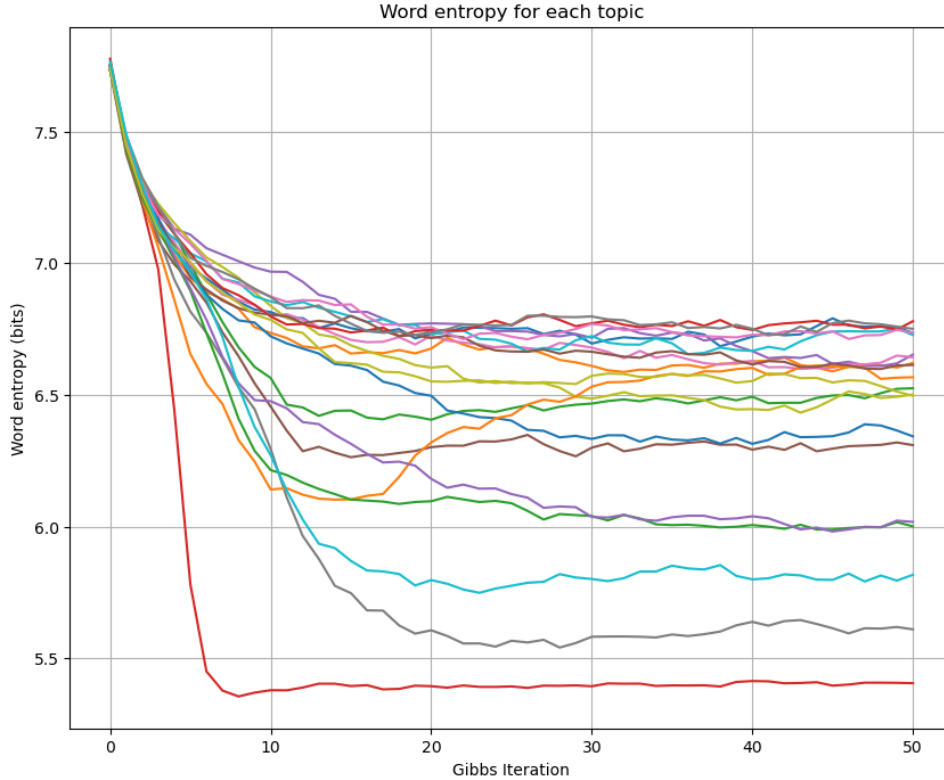


Figure 5: Word entropy (in bits) for each topic wrt Gibbs iterations

Method	Bayesian Inference	Gibbs Sampling	LDA
Perplexity	2684	2100	1701

Table 2: Perplexity (rounded to nearest integer) of all documents using different methods

ments to share a single-word distribution. Quite naturally, we can see that adding more parameters to the model considerably increases its performance, measured using perplexity.

The choice of the number of topics K requires careful analysis of the bias-variance tradeoff. With too few topics, the model cannot capture nuanced thematic variations, resulting in high bias and suboptimal perplexity. Conversely, too many topics introduce excessive parameters, leading to overfitting and topic fragmentation. This manifests as redundant or sparse topics that fail to capture meaningful document-level patterns. The data suggests $K = 20$ may be suboptimal - several topics show very low proportions across documents and fail to capture distinct themes. A lower K value would reduce model complexity while maintaining thematic resolution. This would consolidate related topics and eliminate redundant ones, potentially improving both interpretability and generalization performance. The optimal K could be formally determined through cross-validation of perplexity scores across different K values.